

## TACHEOMETRIC SURVEYING:

1) What is tacheometry?

Tacheometry is the branch of surveying in which horizontal and vertical distances are determined by taking angular observations with an instrument known as a tacheometer.

2) What are the advantages of tacheometric surveying?

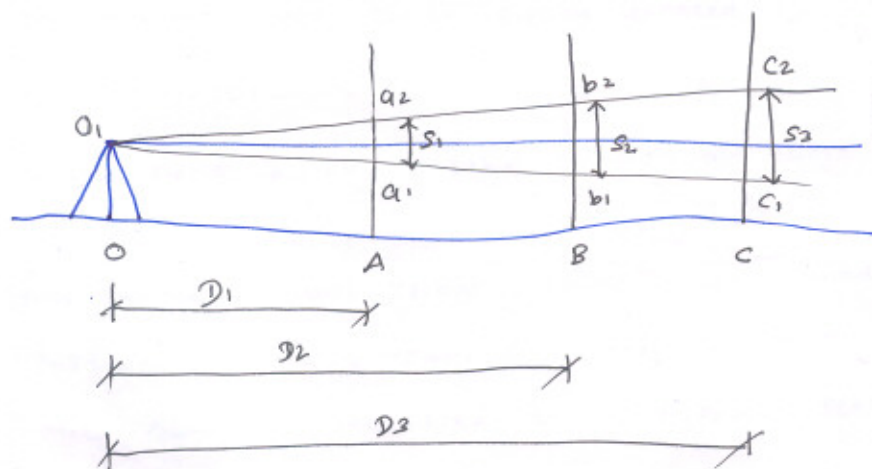
- a) chaining is completely eliminated.
- b) Can be adopted where direct levelling & chaining are not possible.
- c) Very rapid, though not accurate.
- d) Contour map can be prepared easily.

3) What are the instruments used in tacheometry?

- a) Tacheometer [Theodolite + stadia diaphragm]  
+ Anallactic lens
- b) Levelling staff
- c) Stadia rod [for long sights; graduations bold & clear; 0.001 m]

4) What is the principle of tacheometry?

- Isosceles triangle
- Ratio of distance of the base from the apex is constant  
Length of the base

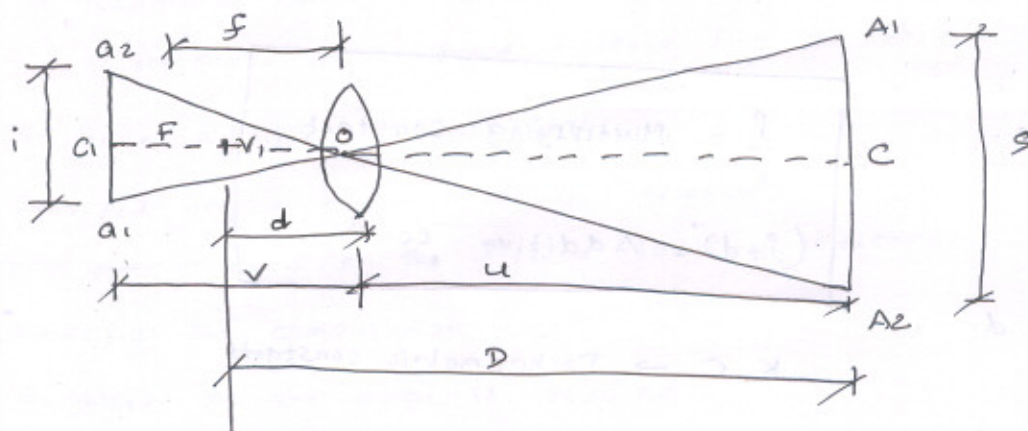


$$\frac{D_1}{s_1} = \frac{D_2}{s_2} = \frac{D_3}{s_3} = \frac{f}{i}$$

f - focal length of the Objective

i - Stadia intercept

5) Derive the distance equation using theory of stadia tachometry.



O - Optical centre of object glass

A<sub>1</sub>, A<sub>2</sub>, C - Readings on staff cut by three hairs

a<sub>1</sub>, a<sub>2</sub>, c - Bottom, top, central hairs of diaphragms

a<sub>1</sub>, a<sub>2</sub> = i = length of image

A<sub>1</sub>, A<sub>2</sub> = S = Staff intercept

F = Focus

V = Vertical axis of instrument

f = Focal length of object glass

d = distance between optical centre and vertical axis of the instrument

u = distance between optical centre and staff.

v = distance between optical centre and image.

From similar triangles,

$$\triangle Oa_1a_2 \sim \triangle OA_1A_2;$$

$$\frac{i}{S} = \frac{v}{u} \Rightarrow \boxed{v = \frac{iu}{S}}$$

From the properties of lenses,

$$\boxed{\frac{1}{v} + \frac{1}{u} = \frac{1}{f}} \Rightarrow \frac{1}{iu/S} + \frac{1}{u} = \frac{1}{f} \Rightarrow \boxed{\frac{S}{iu} + \frac{1}{u} = \frac{1}{f}}$$



$$\frac{1}{u} \left[ \frac{s}{i} + 1 \right] = \frac{1}{f}$$

$$u = \left( \frac{s}{i} + 1 \right) f$$

$$D = u + d$$

$$D = \left( \frac{s}{i} + 1 \right) f + d$$

$$= \frac{s}{i} \times f + f + d$$

$$D = \left( \frac{f}{i} \right) s + (f + d)$$

$$D = Ks + C$$

$\frac{f}{i}$  = Multiplying constant

$(f + d)$  = Additive constants

K, C  $\rightarrow$  Tacheometric constants

6) What are the characteristics of a tacheometer?

- (i) Multiplying constant  $\left( \frac{f}{i} \right)$  should be 100.
- (ii) Telescope should be powerful [magnification 20 to 30 diameters]
- (iii) Aperture of the objective 35 to 45 mm diameter for a bright image
- (iv) Telescope with anallactic lens to make  $C = 0$
- (v) Eye piece should be of greater magnifying power than usual.

NOTE: Anallactic lens  $\rightarrow$   $K = 100$ ;  $C = 0$ ; Normally

7) Determine the tacheometric or stadia constants.

1) Lab measurements  $\left[ \frac{1}{f} = \frac{1}{u} + \frac{1}{v} \right]$

2) Field measurements

$$D_1 = \left( \frac{f}{i} \right) s_1 + (f + d)$$

$$D_2 = \left( \frac{f}{i} \right) s_2 + (f + d)$$

8) What is an anallactic lens?

- An additional convex lens is provided between the eyepiece and the object glass at a fixed distance from the object glass.
- To make additive constant zero.
- Provided only in external focusing telescope.
- Not ~~per~~ required in the internal focusing telescope.
- Simplifies the computation.
- Brightness of the image is reduced.

9) What are the methods of tachometry?

a: Stadia Method

- Fixed Hair Method  
(Very common)
- Moveable hair method  
(Not generally used)

b: Tangential method:

- No Stadia hairs
- Two observations
- Not used

10) Explain Fixed-hair method.

Case ci): Line of sight horizontal; staff held vertical

Case cii): Line of sight inclined; staff held vertical

Case ciii): Line of sight inclined; staff held normal to it.

11) Determine the values of Stadia constants from the following observations.

| Instrument Station | Staff reading on | Distance | Stadia readings |       |
|--------------------|------------------|----------|-----------------|-------|
|                    |                  |          | Lower           | Upper |
| O                  | A                | 150      | 1.255           | 2.750 |
|                    | B                | 200      | 1.000           | 3.000 |
|                    | C                | 250      | 0.750           | 2.255 |



SOLUTION:-

$$D = KS + C.$$

For each case; form the equation

$$150 = (1.495)K + C \rightarrow (1)$$

$$200 = (2.000)K + C \rightarrow (2)$$

$$250 = (2.505)K + C \rightarrow (3)$$

$$\text{Average } K = 99;$$

$$\text{Average } C = 2$$

Solving (1) and (2);  $K = 99; C = 2$

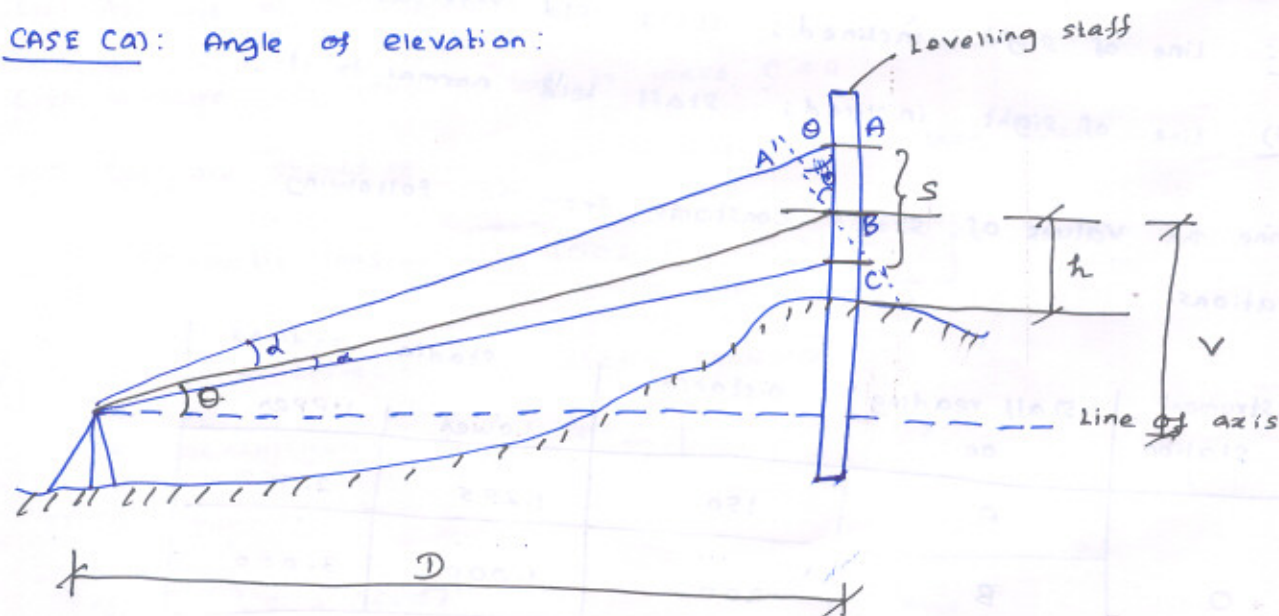
Solving (2) and (3);  $K = 99; C = 2$

Solving (3) and (1);  $K = 99; C = 2$

11) Explain the case "Line of sight inclined" Staff held vertical.

- a) considering the angle of elevation (positive)
- b) considering the angle of depression (negative)
- c) considering both angles of depression (negative)

CASE (a): Angle of elevation:



$T$  = inst. station;  $T_i$  = Axis of instrument;  $P$  = staff station.

$ABC$  = Position of staff cut by hairs.

$S = AC$  = Staff intercept

$h$  = central hair reading.

$V$  = Vertical distance between instrument axis and central hair.

$D$  = Horizontal distance between instrument axis and staff.

$L$  = Inclined distance between instrument axis and  $B$

$\theta$  = Angle of elevation

$\alpha$  = Angle made by outer and inner rays with central ray.

Inclined Distance  $L = KS' + C$

$$= K[S \cos \theta] + C$$

Horizontal Distance  $D = L \cos \theta$

$$= [KS \cos \theta + C] \cos \theta$$

$$D = KS \cos^2 \theta + C \cos \theta$$

$$\angle ABA' = \angle CB'C' = \theta;$$

$$\angle AA'B = 90^\circ + \alpha; \quad \angle BC'C = 90^\circ - \alpha$$

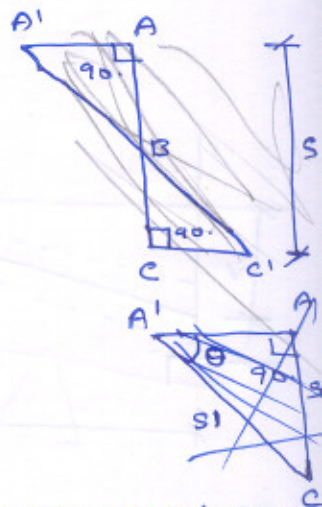
The angle  $\alpha$  is very small;  $\angle AA'B$  and  $\angle BC'C$  may be taken equal to  $90^\circ$ .

$$V = L \sin \theta$$

$$V = \frac{KS \sin 2\theta}{2} + C \sin \theta$$

$$V = D \tan \theta$$

$$RL \text{ of Staff Station } P = RL \text{ of axis of instrument} + V - h$$



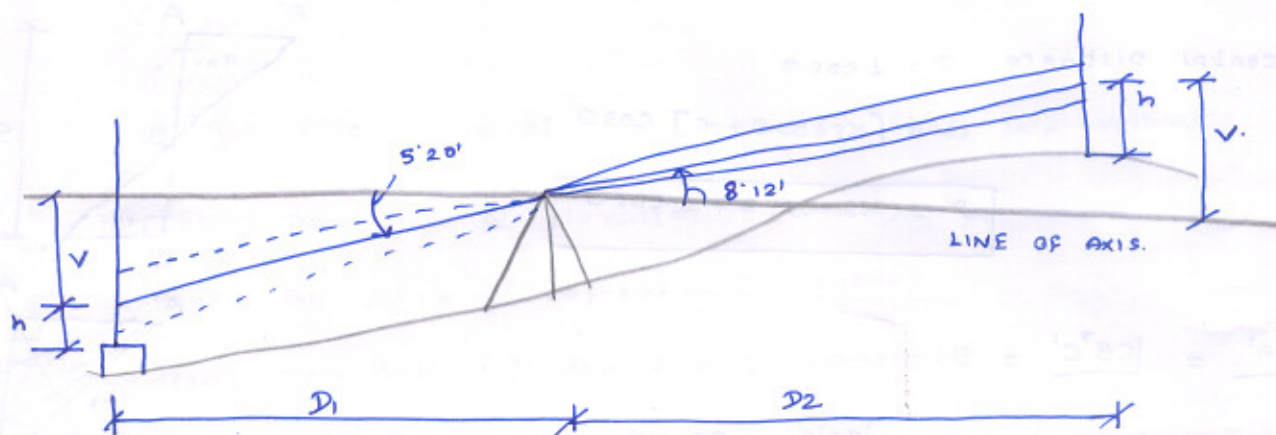


PROBLEMS:-

- 1) A tachometer was set up at a station C and the following readings were obtained on a staff vertically held.

| Inst. Station | Staff Station | Vertical angle | Hair readings       | Remarks    |
|---------------|---------------|----------------|---------------------|------------|
| C             | BM            | -5'20'         | 1.150, 1.800, 2.450 | RL of BM = |
| C             | D             | +8'12'         | 0.750, 1.500, 2.250 | 750.50 m   |

Calculate the horizontal distance CD and RL of D when the constants of instrument are 100 and 0.15

FIRST OBSERVATION:

$$S_1 = 2.450 - 1.150 = 1.300 \text{ m}$$

$$\theta_1 = 5'20' \text{ (depression)}$$

$$V_1 = \frac{K \cdot S_1 \cdot \sin 2\theta}{2} + c \sin \theta = \frac{(100) \times 1.300 \times \sin 2(5'20')}{2} + (0.15) \times \sin 5'20'$$

$$= 12.045 \text{ m.}$$

$$\text{RL of instrument} = \text{RL of B.M} + h_1 + V_1$$

$$= 750.50 + 1.800 + 12.045 = 764.345 \text{ m}$$

SECOND OBSERVATION:

$$S_2 = 2.250 - 0.750 = 1.500 \text{ m}$$

$$\theta_2 = 8'12' \text{ (Elevation)}$$

$$V_2 = \frac{100 \times 1.500 \times \sin 2 \times 8'12'}{2} + 0.15 \times \sin 8'12'$$

$$= 21.947 \text{ m.}$$

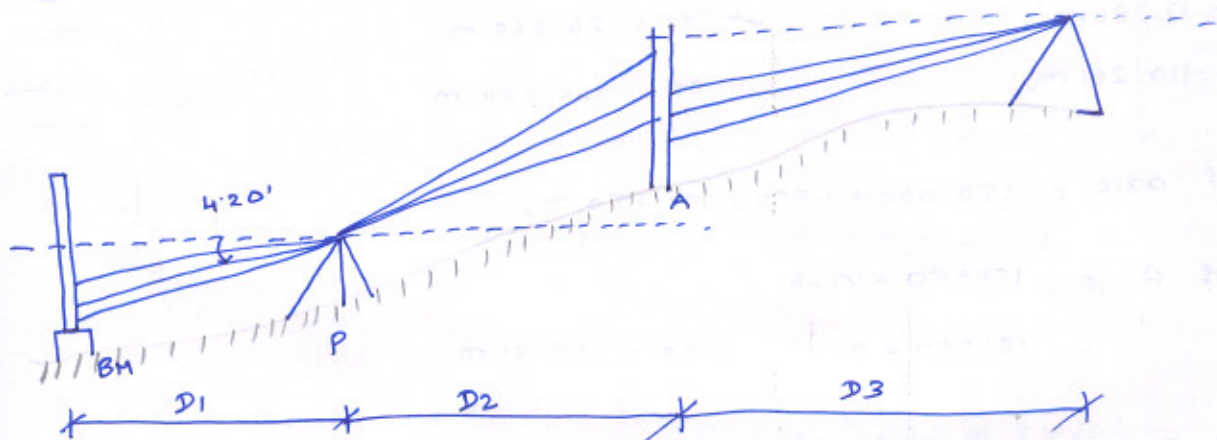
$$RL \text{ of } D = RL \text{ of inst. axis} + V_2 - h_2$$

$$= 764.345 + 21.197 - 1.500 = 784.042 \text{ m.}$$

2) The following observations were taken with a tachometer fitted with an anallactic lens, the staff being vertically held. The constant of the tachometer is 100.

| Inst. Station | H.I   | Staff Station | Vertical angle | Staff readings      | Remarks     |
|---------------|-------|---------------|----------------|---------------------|-------------|
| P             | 1.255 | BM            | -4° 20'        | 1.325, 1.825, 2.325 | RL of B.M = |
| P             | 1.255 | A             | +6° 30'        | 0.850, 1.600, 2.350 | 255.750 m   |
| B             | 1.450 | A             | -7° 24'        | 1.715, 2.315, 2.915 |             |

Calculate RL of B and distance between A and B:



First Observation:

$$V_1 = 100 \left[ \frac{2.325 - 1.325}{2} \right] \times \sin 8' 40'' = 7.534 \text{ m}$$

$$V_2 = 100 \left[ \frac{2.350 - 0.850}{2} \right] \times \sin 13' 0'' = 16.871 \text{ m}$$

$$V_3 = 100 \left[ \frac{2.915 - 1.715}{2} \right] \times \sin 14' 48'' = 15.326 \text{ m}$$

$$RL \text{ of axis when inst at } P = RL \text{ of BM} + h_1 + V_1$$

$$= 255.750 + 1.825 + 7.534 = 265.109 \text{ m.}$$

$$RL \text{ of } A = RL \text{ of inst. axis} + V_2 - h_2 = 265.109 + 16.871 - 1.600 = 280.380 \text{ m.}$$



$$RL \text{ of } B = RL \text{ of } A + h_3 + V_3 - H-I$$

$$= 286.300 + 2.315 + 15.326 - 1.450 = 296.591 \text{ m.}$$

$$\text{Distance between A and B} = 118.009 \text{ m.}$$

$$3) K=100; C=0;$$

| Inst. Station | Height of inst. | Staff Station | WCB     | Vertical angle | Hair Readings       |                     |
|---------------|-----------------|---------------|---------|----------------|---------------------|---------------------|
| O             | 1.550           | A             | 30° 30' | 4° 30'         | 1.155, 1.755, 2.355 | RL of O = 150.000 m |
| O             | 1.550           | B             | 75° 30' | 10° 15'        | 1.250, 2.000, 2.750 |                     |

Calculate the distance AB and RLs of A and B. Find also the gradient of the line AB.

SOLUTION:

First observation: [OA]

$$V_1 = 9.386 \text{ m;}$$

$$D_1 = 119.261 \text{ m;}$$

$$RL \text{ of axis} = 150.000 + 1.550 = 151.550 \text{ m;}$$

$$\begin{aligned} RL \text{ of } A &= 151.550 + V_1 - h_1 \\ &= 151.550 + 9.386 - 1.755 = 159.181 \text{ m} \end{aligned}$$

$$\begin{aligned} RL \text{ of } B &= 151.550 + V_2 - h_2 \\ &= 151.550 + 26.265 - 2.000 = 175.815 \text{ m} \end{aligned}$$

$$OA = D_1 = 119.261 \text{ m}$$

$$OB = D_2 = 145.250 \text{ m}$$

$$\angle = 75^\circ 30' - 30^\circ 30' = 45^\circ 00'$$

$$\begin{aligned} AB &= \sqrt{OA^2 + OB^2 - 2 \times OA \times OB \times \cos 45^\circ} \\ &= 104.05 \text{ m.} \end{aligned}$$

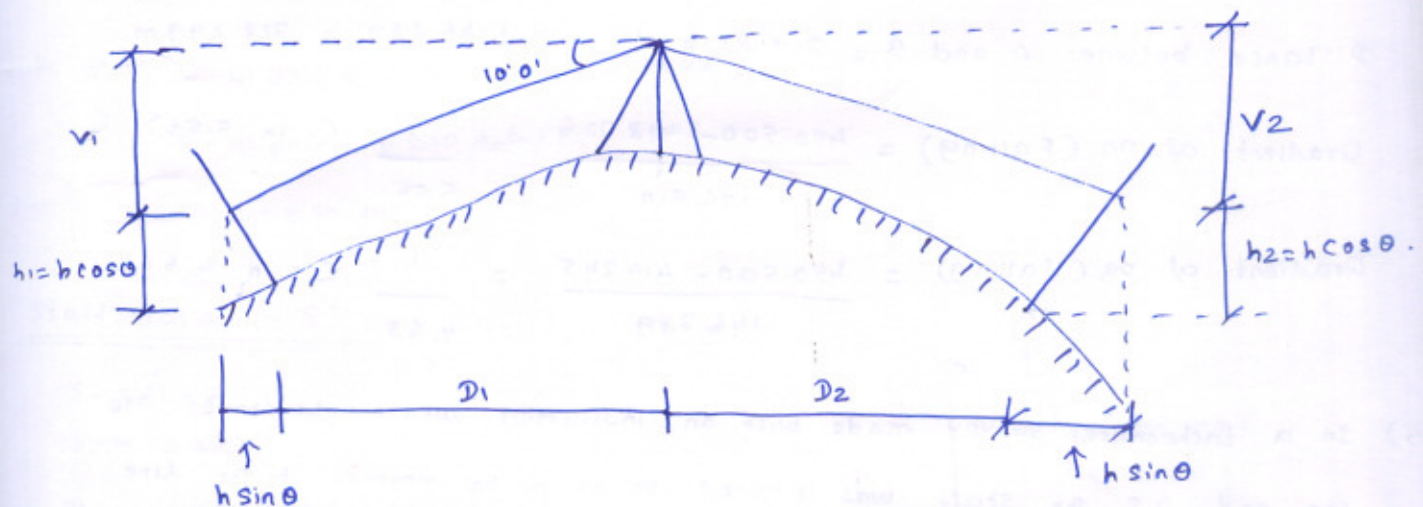
$$\text{Gradient of } AB = \frac{\text{Difference in level}}{\text{Hor. Distance}} = \frac{16.634}{104.05} = \frac{1}{6.25} \quad \left( \underline{\underline{1 \text{ in } 6.25}} \right)$$

- 4) Two points A and B are on opposite sides of a summit. The tacheometer was set up at P on top of the summit, and the following readings were taken.

| Inst. Station | Height of instrument | Stadd Station | Vertical angle  | Hair readings       | Remark              |
|---------------|----------------------|---------------|-----------------|---------------------|---------------------|
| P             | 1.500                | A             | $-10^{\circ}0'$ | 1.150 2.050 2.950   | RL of P = 450.500 m |
| P             | 1.500                | B             | $-12^{\circ}0'$ | 0.855, 1.605, 2.355 |                     |

The tacheometer is fitted with an anallactic lens, the multiplying constant being 100. The staff was held normal to the line of sight.

- Find (a) The distance between A and B and  
(b) The gradients of lines PA and PB.



$$\left. \begin{aligned} V_1 &= Ks \sin \theta + C \sin \theta \quad [V = L \sin \theta] \\ D_1 &= L \cos \theta + h \sin \theta \end{aligned} \right\} \text{For Angle of Elevation.}$$

$$D = L \cos \theta - h \sin \theta = (Ks + C) \cos \theta - h \sin \theta = Ks \cos \theta + C \cos \theta - h \sin \theta$$

$$V = L \sin \theta = Ks \sin \theta + C \sin \theta;$$

SOLUTION:

$$V_1 = Ks \sin \theta_1 = 100 \times (2.950 - 1.150) \times \sin 10^{\circ} = 31.256 \text{ m}$$

$$h_1 = h \cos \theta = 2.050 \times \cos 10^{\circ} = 2.018 \text{ m}$$

$$V_2 = Ks \sin \theta_2 = 100 \times (2.355 - 0.855) \times \sin 12^{\circ} = 31.186 \text{ m}$$

$$h_2 = 1.605 \times \cos 12^{\circ} = 1.569 \text{ m}$$



$$RL \text{ of "A" instrument axis} = 450 + 1.500 = 452.000 \text{ m}$$

$$RL \text{ of A} = RL \text{ of inst. axis} - V_1 - h_1$$

$$= 452.000 - 31.256 - 2.018$$

$$= 418.726 \text{ m.}$$

$$RL \text{ of B} = 452.00 - V_2 - h_2$$

$$= 452.00 - 31.186 - 1.569$$

$$= 419.245 \text{ m}$$

$$D_1 = 100 \times (2.950 - 1.150) \cos 10^\circ - 2.050 \times \sin 10^\circ$$

$$= 177.265 - 0.355 \Rightarrow 176.91 \text{ m}$$

$$D_2 = 100 \times (2.355 - 0.855) \times \cos 12^\circ - 1.605 \times \sin 10^\circ$$

$$= 146.722 - 0.333 \Rightarrow 146.389 \text{ m.}$$

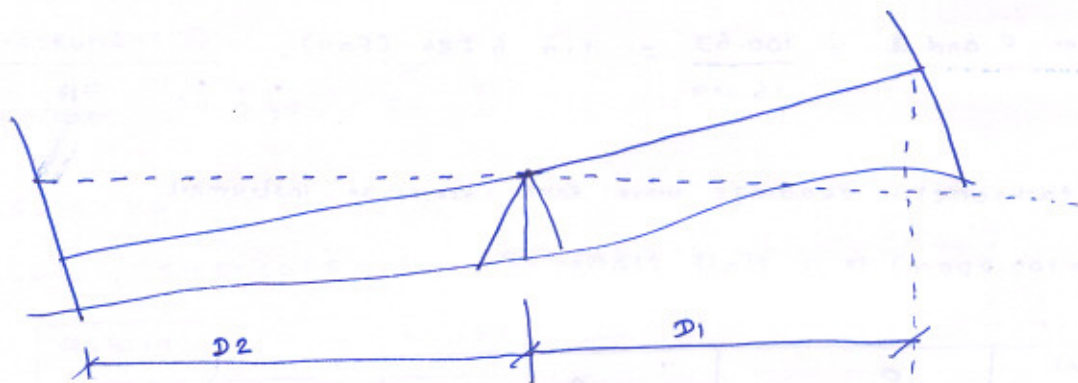
$$\text{Distance between A and B} = D_1 + D_2 = 176.910 + 146.389 = 323.299 \text{ m}$$

$$\text{Gradient of PA (Falling)} = \frac{450.500 - 418.726}{176.910} = \frac{1}{5.56} \quad (\text{i in 5.56})$$

$$\text{Gradient of PB (falling)} = \frac{450.500 - 419.245}{146.389} = \frac{1}{4.68} \quad (\text{i in 4.68})$$

- 5) In a tacheometer survey made with an instrument whose constants are 100 and 0.5, the staff was inclined so as to be normal to the line of sight for each reading. Two sets of readings were as given below. Calculate the gradient between the staff stations P and Q and the reduced level of each if that of R is 41.800 m.

| Inst. Station | Height of inst. axis | Staff Station | Bearing | Vertical angle | Stadia Reading      |
|---------------|----------------------|---------------|---------|----------------|---------------------|
| R             | 1.600                | P             | 85°     | +4°30'         | 1.000, 1.417, 1.833 |
|               |                      | Q             | 135°    | -4°00'         | 1.000, 1.657, 2.313 |



Staff station @ P :-

$$S_1 = 0.833 \text{ m};$$

$$h_1 = 1.413 \text{ m};$$

$$D_1 = (Ks \cos \theta + c \cos \theta + h \sin \theta)$$

$$= 83.65 \text{ m};$$

$$V_1 = Ks \sin \theta + c \sin \theta$$

$$= 6.575 \text{ m}.$$

$$RL \text{ of } P = 41.800 + 1.600 + 6.575 - h \cos \theta$$

$$= 41.800 + 1.600 + 6.575 - 1.413$$

$$= 48.562 \text{ m}$$

Staff Station @ Q:

$$S_2 = 1.313$$

$$h_2 = 1.657$$

$$D_2 = (Ks \cos \theta + c \cos \theta) - h \sin \theta$$

$$= 131.36 \text{ m}$$

$$V_2 = 9.194 \text{ m}$$

$$RL \text{ of } Q = 41.800 + 1.600 - 9.194 - 1.657 \times \cos 50^\circ$$

$$= 32.553 \text{ m}.$$

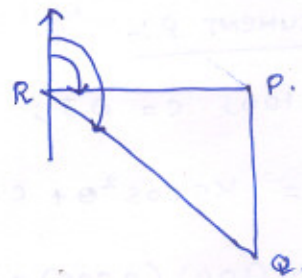
$$\text{Diff. in level } P \text{ and } Q = 48.562 - 32.553 = 16.009 \text{ m}$$

$$\text{Angle } PRQ = 50^\circ$$

Applying cosine formula to  $\Delta RPQ$ , we get

$$PQ^2 = (83.65)^2 + (131.36)^2 - 2 \times 83.65 \times 131.36 \times \cos 50^\circ$$

$$= 10126.520 \Rightarrow \boxed{PQ = 100.63 \text{ m}}$$





Gradient between P and Q =  $\frac{100.63}{16.009} = 1 \text{ in } 6.286 \text{ (Fall)}$

- 6) Two sets of tacheometric readings were taken from an instrument Station A (RL = 100.000m) to a staff station B.

| Instrument |    |    | P              | Q                       |
|------------|----|----|----------------|-------------------------|
| K          |    |    | 100            | 95                      |
| C          |    |    | 30 cm          | 45 cm                   |
| H.I        |    |    | 1.40 m         | 1.45 m                  |
| Staff held |    |    | Vertical       | Normal to line of sight |
| Inst.      | at | to | Vertical angle | Stadia readings         |
| P          | A  | B  | 5° 44'         | 1.090, 1.440, 1.795     |
| Q          | A  | B  | 5° 44'         | ?                       |

- Determine (a) The distance between inst. station and staff station  
(b) RL of Staff Station B  
(c) The stadia readings with inst. Q.

SOLUTION:

INSTRUMENT P:-

$$K=100; C=0.3; S=1.795-1.090=0.705 \text{ m};$$

$$D_I = K S \cos^2 \theta + C \cos \theta$$

$$= (100)(0.705) \times \cos^2 (5^\circ 44') + (0.3) \times \cos 5^\circ 44'$$

$$D_I = 70.095 \text{ m}$$

$$V_I = \frac{K S \sin 2\theta}{2} + C \sin \theta$$

$$= (100)(0.705) \times \sin 2 \times (5^\circ 44') + 0.3 \times \sin (5^\circ 44')$$

$$V_I = 7.038 \text{ m}$$

$$\text{RL of staff station} = \text{RL of A} + \text{H.I} + V - h$$

$$= 100 + 1.400 + 7.038 - 1.440 \Rightarrow 106.998 \text{ m}$$

INSTRUMENT Q:

$$K = \frac{95}{100}; C = 0.45; S = ?$$

$$D_2 = K S \cos \theta + C \cos \theta + h \sin \theta$$

$$70.095 = 95 \times S \times \cos 5'44'' + 0.45 \times \cos 5'44'' + h \times \sin (5'44'')$$

$$\Rightarrow 946.1962 S + h = 697.167 \rightarrow \text{ci)}$$

$$V = K S \sin \theta + C \sin \theta$$

$$= (95 \times S) \sin 5'44'' + 0.45 \times \sin (5'44'')$$

$$= (95S + 0.45) \times 0.0999$$

$$\therefore RL \text{ of } B = RL \text{ of } A + H.I + V - h \cos \theta$$

$$= 100 + 1.450 + (95S + 0.45) \times 0.0999 - h \times 0.995$$

$$\Rightarrow 9.5382 S - h = 5.5307 \rightarrow \text{cii)}$$

Solving ci) and cii)

$$S = 0.735244 \text{ m}$$

$$h = 1.4819 \text{ m}$$

$$\text{Stadia upper reading} = 1.4819 + \left( \frac{0.7352}{2} \right) = 1.8495$$

$$\text{Stadia lower reading} = 1.4819 - \left( \frac{0.7352}{2} \right) = 1.1143$$

$$\text{Stadia Readings} = 1.8495, 1.4819, 1.1143$$



## THE MOVEABLE-HAIR METHOD

- Staff intercept is kept constant.
- Distance between the stadia wires is variable
- The staff is provided with two targets a known distance apart and third target in the middle.
- Theodolite + special diaphragm  $\rightarrow$  Subtense theodolite
- Central wire  $\rightarrow$  fixed;  
upper, lower cross hairs — adjusted using micrometer screws.
- Distance of the stadia wires — no. of turns in micrometer screws.
- Fractions — measured on the drum kept on top and bottom of eye-piece.

Line of sight horizontal:

$$D = \frac{K_m \times S}{n} + C_m$$

$K_m$  — constant varying from (600–1000)

$n$  — Sum of the readings in the micrometer

$S$  — Staff intercept (distance between targets)

Line of sight inclined:

$$D = \frac{K_m \times S \times \cos^2 \theta}{n} + C_m \cos \theta$$

1) The micrometer readings of a subtense theodolite are 3.455 and 3.405. The distance between the targets is 3m.  $K_m = 600$ ;  $C_m = 0.5$ ;

$$D = \frac{K_m \times S}{n} + C_m$$

$$= \frac{600 \times 3}{6.860} + 0.5$$

$$= 262.89 \text{ m}$$

- 2). The same distance was measured by a tacheometer and a subtense theodolite.

$$S = 1.255; \quad \theta = 5^\circ 0' \quad C = 0.2; \quad K = 100 \quad [\text{Tacheometer}]$$

$$S = 2\text{ m}; \quad \theta = 5^\circ 30' \quad K = 1000; \quad C_m = 0.3 \quad [\text{Subtense theodolite}]$$

$$1) \quad D = K S \cos^2 \theta + C \cos \theta$$

$$= (100) (1.255) \times \cos^2 (5^\circ) + 0.2 \times \cos 5^\circ 00'$$

$$= 124.75 \text{ m}$$

$$II) \quad D = \frac{K_m \times S \times \cos^2 \theta}{n} + C_m \cos \theta$$

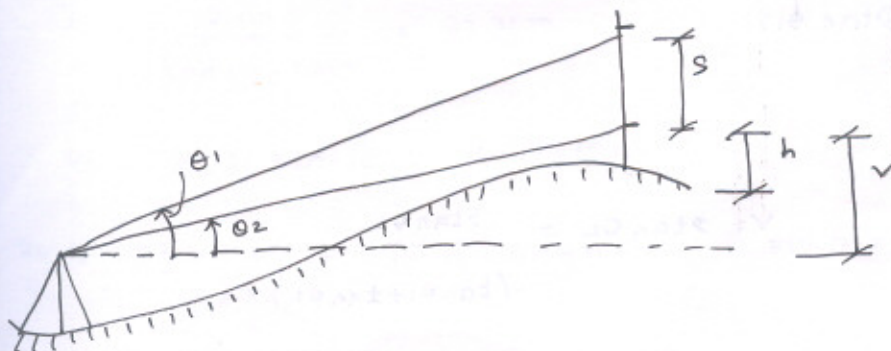
$$124.75 = \frac{1000 \times 2 \text{ m} \times \cos^2 5^\circ 30'}{n} + 0.3 \times \cos 5^\circ 30'$$

$$n = 15.91$$

### TANGENTIAL METHOD OF TACHEOMETRY:

- Method is used when the theodolite is a simple transit type
- Does not carry a stadia diaphragm
- Staff consists of two vanes

- a) Both angles are angles of elevation:



$$D \tan \theta_1 = (V + S)$$

$$V = D \tan \theta_2$$

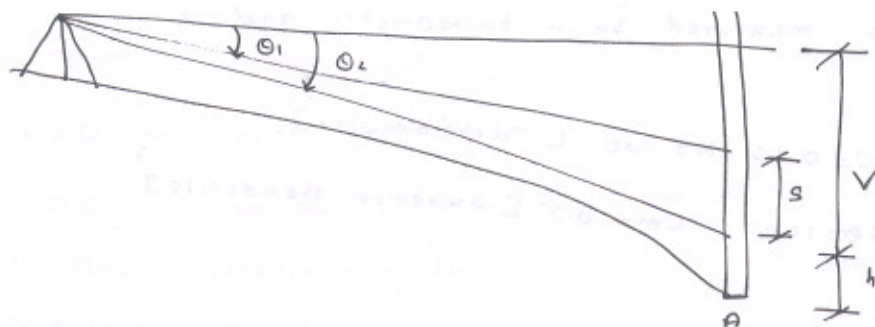
$$S = D (\tan \theta_2 - \tan \theta_1)$$

$$D = \frac{S}{\tan \theta_2 - \tan \theta_1}$$

$$V = \left( \frac{S}{\tan \theta_2 - \tan \theta_1} \right) \cdot \tan \theta_2$$

$$\text{RL of A} = \text{RL of inst. axis} + V - h$$





$$V = D \tan \theta_2;$$

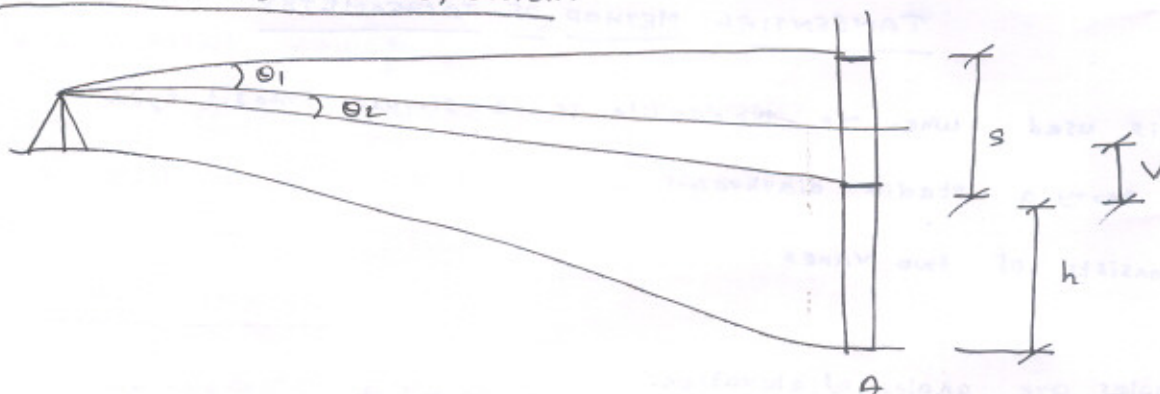
$$V - S = D \tan \theta_1;$$

$$S = D (\tan \theta_2 - \tan \theta_1)$$

$$D = \frac{S}{(\tan \theta_2 - \tan \theta_1)}; \quad V = D \tan \theta_2 = \frac{S \tan \theta_2}{(\tan \theta_2 - \tan \theta_1)}$$

$$\text{RL of A} = \text{RL of inst. axis} - V - h$$

3) One elevation; one depression:



$$V = D \tan \theta_2; \quad S - V = D \tan \theta_1;$$

$$S = D (\tan \theta_1 + \tan \theta_2)$$

$$D = \frac{S}{\tan \theta_1 + \tan \theta_2}$$

$$V = D \tan \theta_2 = \frac{S \tan \theta_2}{(\tan \theta_1 + \tan \theta_2)}$$

$$\text{RL of A} = \text{RL of inst. axis} - V - h$$

1) To find the RL of Station B, two observations are taken by a theodolite from station A, one to a BM, other to station B. The records are as follows.

| Inst. Station | Staff position | Target | Vertical angle   | Staff reading |                      |
|---------------|----------------|--------|------------------|---------------|----------------------|
| A             | BM             | lower  | $-10^{\circ}00'$ | 0.655         | RL of BM = 510.500 m |
|               |                | upper  | $-7^{\circ}00'$  | 2.655         |                      |
| A             | B              | lower  | $-5^{\circ}00'$  | 1.250         |                      |
|               |                | upper  | $+4^{\circ}00'$  | 3.200         |                      |

Det. RL of B; Distance between BM and B.

SOLUTION:

First observation:

$$S_1 = 2.000 \text{ m}; h_1 = 0.655; V_1 = D \tan 10^{\circ}$$

$$V_1 - S_1 = D \tan 7^{\circ}$$

$$S_1 = D (\tan 10^{\circ} - \tan 7^{\circ})$$

$$D = \frac{2.000}{\tan 10^{\circ} - \tan 7^{\circ}} = 37.31 \text{ m.}$$

$$V_1 = 37.31 \times \tan 10^{\circ} = 6.578 \text{ m.}$$

$$\text{RL of instrument axis} = \text{RL of BM} + h_1 + V_1 = 517.733 \text{ m.}$$

Second observation:

$$V_2 = D \tan 5^{\circ} \quad S_2 = 1.950 \text{ m.}$$

$$S_2 - V_2 = D \tan 4^{\circ}$$

$$D = \frac{1.950}{(\tan 5^{\circ} + \tan 4^{\circ})} = 12.396 \text{ m}$$

$$V_2 = D \tan 5^{\circ} = 1.085 \text{ m.}$$

$$\text{RL of B} = 515.398 \text{ m.}$$

Distance between BM and B =

$$37.31 + 12.396 = 49.706 \text{ m}$$